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THE UNIVERSITY OF HONG KONG

STAT7008 Programming for data science  
Group Project Report

*Project 11: A collaborative workspace system*

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Please note that we want to have an individual assessment

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GitHub repository: <https://github.com/ZiyuWang1121/Collaborative-Workspace-System>

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## **1. Introduction**

### **1.1 Problem Statement**

Effective project management in the IT industry is challenged by manual and outdated task assignment methods. Traditional approaches often fail to align nuanced skill requirements from task descriptions with employee expertise, experience, and workload (Masood Z, 2021). Rapid technological evolution further widens the gap between static skill inventories and dynamic project needs, leading to skill mismatches, employee overload, and project inefficiencies. A lack of integrated, data-driven tools for automated and predictive task-employee matching exacerbates these issues.

### **1.2 Proposed Solution**

This project proposes an intelligent collaborative workspace that automates task allocation in IT projects. By leveraging natural language processing (NLP) and machine learning, the system extracts skill requirements from tasks and matches them with employee profiles using simulated job and skills data. The solution aims to reduce assignment errors, optimize resource utilization, and improve project outcomes through data-driven matching.

### **1.3 Project Aims and Objectives**

This project aims to develop a system that automates task-employee matching in IT projects. The system supports core project management functions, including project and task definition, scheduling, role profiling, progress tracking, and basic notifications. The corresponding objectives are as follows:

- (1) Design the databases, identify/collect/simulate the data, and draw object-oriented diagrams for the system's functions.
- (2) Develop all these functions (i.e., a library's function call), and group some functions into a class (i.e, a program library).
- (3) Use a data mining model for relating the project skill requirements and staff profiles.
- (4) Use data visualization to generate and visualize the project skills requirement and staff profiles analysis.

## **1.4 Scope of Study**

This study focuses on developing a system for automated task-employee matching in simulated IT project environments. It includes NLP-based skill extraction, matching algorithm development, core system design, and basic visualization. Excluded are real-time enterprise integration, full-scale deployment, advanced collaboration features, and validation beyond simulated IT domains.

## **2. Literature Review**

Effective task-employee matching in IT projects is increasingly dependent on automated, data-driven approaches. As project complexity grows, the manual mapping of developer competencies to technical requirements becomes a bottleneck, prone to cognitive bias and inefficiency. This review synthesizes key research across three critical areas: (1) the use of Natural Language Processing (NLP) for skill extraction, (2) the application of machine learning for matching and prediction, and (3) their integration within IT project management systems. Finally, it identifies existing research gaps that this project aims to address.

### **2.1 Skill Extraction from Unstructured Text Using NLP**

The automation of matching begins with the accurate identification of skills from unstructured text, such as job descriptions, resumes, and technical task tickets. Historically, this process relied on keyword-based methods like Term Frequency-Inverse Document Frequency (TF-IDF) or simple "bag-of-words" models. However, recent literature highlights the severe limitations of these approaches, particularly their inability to handle polysemy (words with multiple meanings) and synonymy (different words for the same concept) in dynamic IT vocabularies.

Current research demonstrates a decisive shift toward context-aware embedding models and Large Language Models (LLMs). Senger et al. (2024), in a comprehensive survey of NLP for Human Resources, note that transformer-based architectures—such as BERT and RoBERTa—have effectively replaced static keyword parsers. These models generate dense vector representations where semantically related terms (e.g., "React.js" and "Front-end Development") are positioned closely in vector space, even if they never appear together in the same document.

Validating this shift, Zhang et al. (2024) developed "ResuFormer," a transformer-based framework specifically designed for parsing resume blocks. Their study found that while traditional TF-IDF models struggled with the fragmented syntax of bullet points, attention mechanisms allowed ResuFormer to capture long-range dependencies, achieving an F1-score of over 92% in entity recognition tasks. Similarly, Li et al. (2023) introduced "SkillGPT," a system leveraging LLMs to standardize non-standard skill terms into a canonical ontology.

Their results indicate that LLM-based extraction outperforms rule-based systems by a significant margin (15-20% improvement in recall), primarily due to the model's pre-trained knowledge of the IT domain.

These findings validate this project's decision to employ advanced NLP techniques rather than simple keyword matching. By parsing task descriptions into structured skill vectors, the system can ensure that a request for a "Django expert" is mathematically understood as a request for "Python proficiency," a nuance that simple Boolean logic would miss (Herandi et al., 2024).

## **2.2 Matching Algorithms and Predictive Modeling**

Once skills are quantified into vectors, the core challenge shifts to designing matching algorithms that can accurately align these profiles with project requirements. The literature reveals a progression from simple similarity metrics to complex, graph-based predictive modeling.

Early automated matching systems relied heavily on Cosine Similarity to measure the angle between a candidate's skill vector and a job's requirement vector. While computationally efficient, this approach treats skills as isolated attributes. Recent advancements argue for a relational approach. Frazzetto and Haq (2024) propose the use of Graph Neural Networks (GNNs) for candidate-job matching. By modeling the data as a graph—where candidates, skills, and past projects are nodes connected by edges—GNNs can infer "implicit" qualifications. For example, if a developer has successfully completed three projects connected to "Cloud Migration," the GNN infers a high suitability for AWS tasks, even if "AWS" is not explicitly listed in their current skill text (Frazzetto & Haq, 2024).

Beyond merely finding a "match," the next frontier is predicting the outcome of that match. Costa et al. (2022) highlight that modern systems are moving from descriptive matching (who fits?) to predictive analytics (who will succeed?). This distinction is critical for risk mitigation in IT projects. Predictive models now utilize historical data—such as past sprint velocity, bug rates, and team collaboration scores—to forecast the probability of task completion for a specific team composition.

Yu et al. (2025) recently advanced this field with their "CONFIT V2" framework, which utilizes hypothetical resume generation to train matching encoders. Their research demonstrates that training models on outcomes (e.g., whether a candidate was actually hired and performed well) rather than just surface-level text similarity drastically reduces turnover risk. Furthermore, Gan et al. (2024) explored the use of LLM agents to simulate recruitment decisions, finding that predictive agents could identify "high-potential" candidates who lacked perfect keyword matches but possessed the requisite problem-solving patterns.

## 2.3 Integration in IT Project Management Systems

The integration of advanced artificial intelligence (AI) techniques into IT project management systems represents a transformative shift in how agile teams operate, moving beyond static planning to dynamic, intelligent orchestration. While traditional project management tools like JIRA or Asana rely heavily on manual updates and human intuition for task distribution, the next frontier involves systems that actively perceive, analyze, and optimize workflows in real-time. This evolution is driven by two primary technological pillars: Reinforcement Learning (RL) for dynamic task allocation and Large Language Models (LLMs) for semantic talent matching.

Research demonstrates RL's effectiveness in adaptive scheduling. Gayathri et al. (2025) pioneered a hybrid framework combining NLP to parse tasks with RL agents to optimize assignments in real-time, improving sprint velocity and workload balance over traditional methods. Complementing this, Gohar (2024) emphasizes the need for "human-aware" RL systems that account for factors like developer burnout, advocating for "human-in-the-loop" AI where algorithms provide decision support rather than autonomous commands.

On the talent side, LLMs enhance matching precision. Kadam et al. (2025) showed that LLMs (e.g., GPT variants) outperform keyword-based systems by understanding latent skills (e.g., linking "Django" to "Python"). However, Chhatre & Singh (2025) warn of algorithmic bias, stressing that LLM-driven tools require fairness constraints to mitigate risks of discrimination in recruitment.

Despite these advances, a significant fragmentation persists. Taboada et al. (2023) notes that most AI solutions are isolated "point solutions." For example, skill data from LLM-powered hiring (Kadam et al.) is not utilized by RL schedulers (Gayathri et al.) during sprint planning, breaking the project lifecycle continuity.

This project explicitly targets this fragmentation. By proposing a unified system that bridges the gap between semantic skill analysis and dynamic, reinforcement-learning-based scheduling, this research aims to create a continuous feedback loop. In this proposed model, task completion data feeds back into the skill profile, constantly refining the semantic matching algorithm—a feature largely absent in current commercial and academic systems.

## 2.4 Identified Research Gap and Project Contribution

Despite these advancements, a notable gap persists in the literature. Most current prototypes, such as those by Zhang et al. (2024) and Yu et al. (2025), focus exclusively on the recruitment phase (hiring new employees) rather than the internal allocation phase (assigning tasks to existing teams). There is a lack of research on how "hiring-grade" matching algorithms can be adapted for the high-velocity, day-to-day scheduling required in Agile project management. This project aims to bridge this gap by applying these advanced

extraction and predictive matching techniques directly to the internal project lifecycle, optimizing how existing resources are deployed against backlog items.

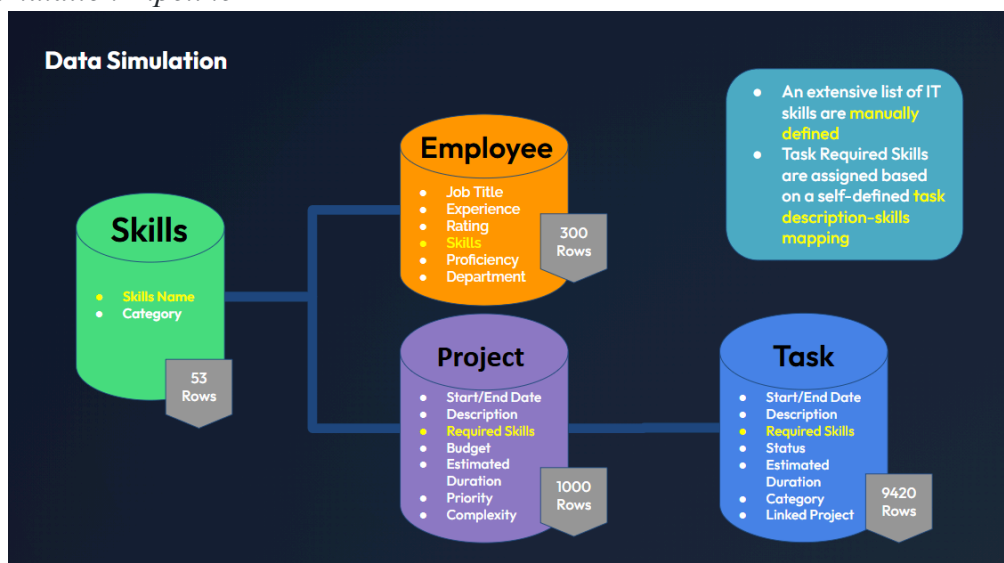
In summary, while existing literature provides robust methods for discrete components like skill extraction and algorithmic matching, integrated, end-to-end prototype systems are scarce. Few studies combine automated matching with a full suite of project management functionalities within a unified, object-oriented framework, especially one that is accessible for demonstration and validation using open datasets. This project contributes by developing a cohesive, intelligent collaborative workspace prototype. It synthesizes established NLP and ML techniques into a functional system that not only matches tasks to employees but also embeds this capability within core project management workflows, thereby addressing the practical gap between algorithmic research and deployable tool development for IT project environments.

### 3. Research Methods

#### 3.1 Data Sources, Data Cleansing, and Pre-processing

To ensure rigor while circumventing privacy constraints, this study generated synthetic data using Python's Faker, NumPy, and Pandas libraries. Four interconnected datasets simulate organizational operations: a skills catalog (52 skills), an employee dataset (300 records), a project dataset (1000 entries), and a task dataset (9420 records). The simulation yielded a clean dataset, with 985 task records (10.5%) intentionally left unassigned to represent legitimate pending work. Validation confirmed data integrity, including unique identifiers, standardized formats, and referential consistency across entities. Temporal columns were converted for analysis, and outlier detection flagged only 0.1% of project budgets. This established a reliable analytical foundation. Below is an illustration of the simulation process:

**Figure 1**  
*Data Simulation Pipeline*

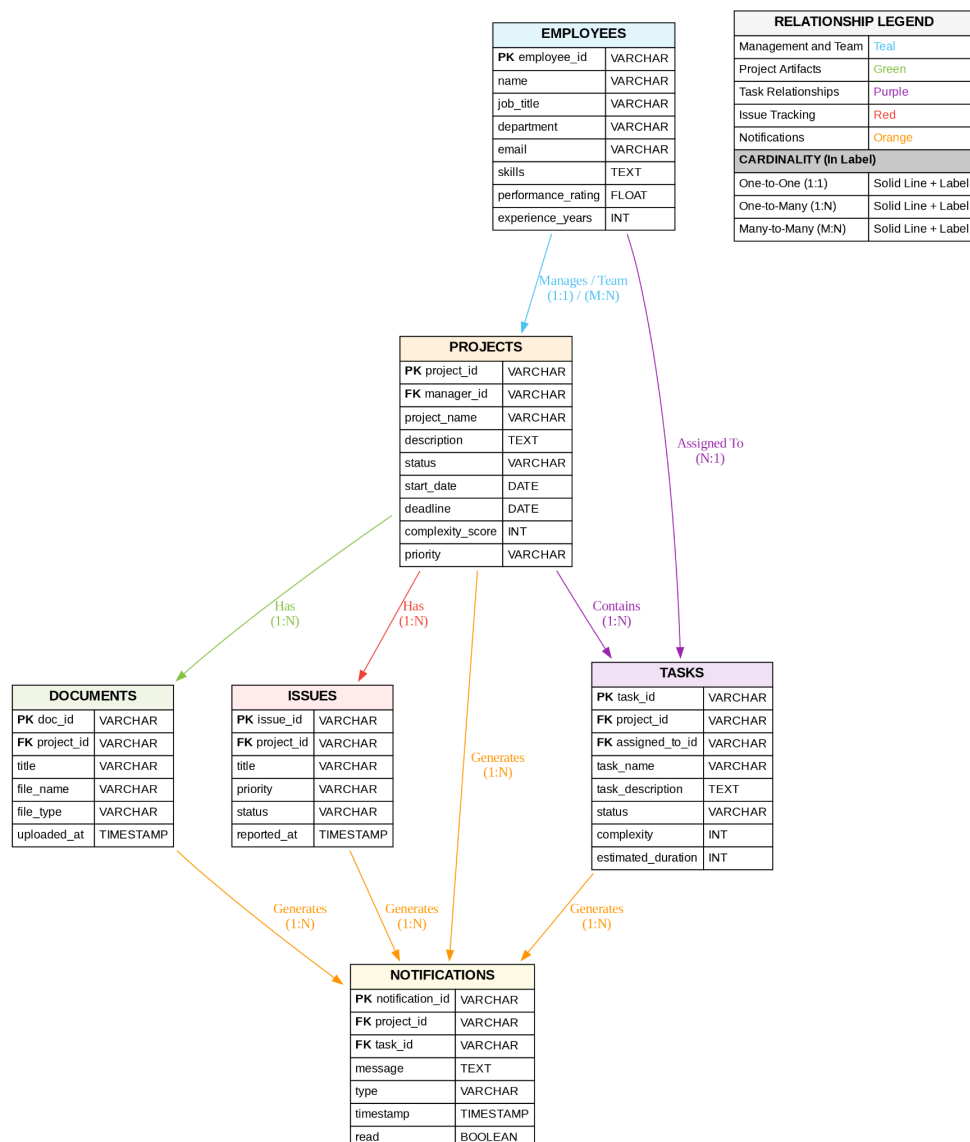


## 3.2 Object-Oriented Programming Design

### Data Schema Design

The system implements a normalized relational schema with six core entities—Employees, Projects, Tasks, Documents, Issues, and Notifications—ensuring data integrity and full auditability. Employees form the central entity, linked to Projects (management relationship) and Tasks (assignment relationship). Supporting entities handle project artifacts (Documents, Issues) and system notifications, with explicitly defined relationships color-coded for clarity (management, task links, artifacts, notifications). This design eliminates redundancy and supports scalability.

**Figure 2**  
*Data Schema*

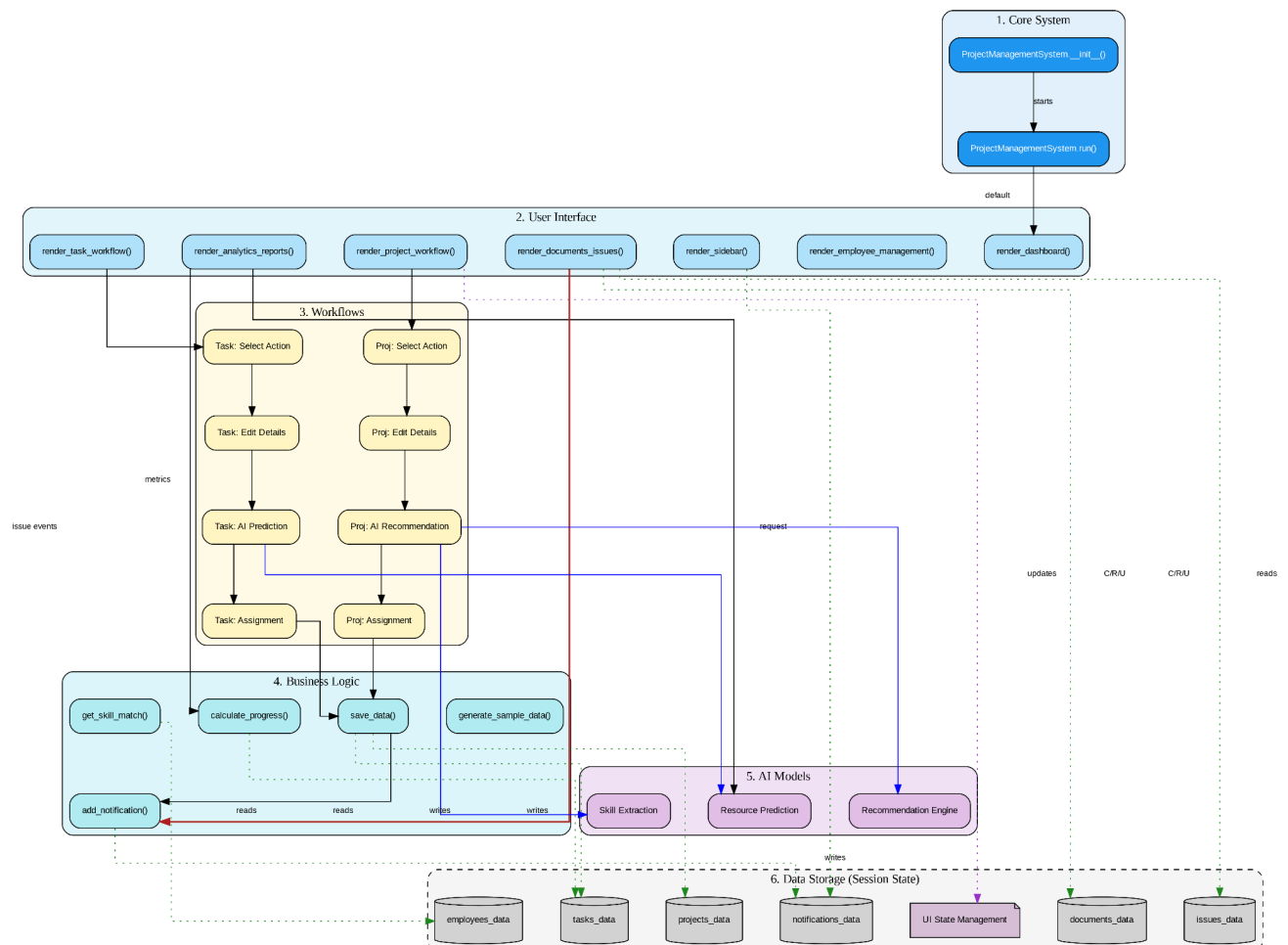




## System Architecture & Function Design

Operational logic is architected as an eight-layer functional hierarchy for modularity. It begins with Core System Initialization, which boots the application. The Main User Interface layer provides six primary UI modules (e.g., dashboards, project/task workflows). User actions trigger vertical workflows: the Project Workflow and Task Workflow layers each guide a five-step process (Select Action, Edit Details, AI Recommendations, Team Assignment, Completion). These interact with the Core Business Logic layer (e.g., `calculate_project_progress()`). An External Model Interfaces layer abstracts calls to the External AI/ML Models layer (Skills Extraction, Task Estimation models), allowing seamless model swapping. Data is managed by the Data Storage layer, supported by an External Libraries layer (Streamlit, Pandas). This design ensures scalability and clean separation of concerns.

**Figure 3**  
*System Function Calls*



### **3.3 Model Design**

#### **Descriptive Analytics, Input and Output Variables**

The analytical framework begins with an analysis that examines distributions and relationships across four domains: employees (experience, performance, skills), projects (complexity, budget, priority), tasks (complexity, duration, status), and a standardized skills catalog. These variables represent the organization's human capital and operational workload.

#### **Models Design**

##### **Part 1: Skill Extraction and Prediction**

This phase automatically extracts required skill sets from project descriptions. The system employs a hybrid approach: initial rule-based text mining for data preparation, followed by a high-accuracy Transformer-based multi-label classifier (e.g., DistilBERT) for prediction. The model encodes descriptions and uses a binary cross-entropy loss to predict relevant skills, with final results determined by a configurable threshold.

##### **Part 2: Team Selection and Optimization**

This stage selects the best-suited team members based on identified requirements and constraints. The algorithm combines a multi-factor scoring model with greedy heuristic optimization. Candidates are scored using weighted factors like skill match ratio and average proficiency, then iteratively selected to maximize coverage of unmet skills, ensuring optimal team composition within size limits.

##### **Part 3: Task Budget and Duration Estimation (Statistics / Regression)**

This model selects the optimal team. The algorithm combines a Multi-Factor Scoring Model (weighting skill match ratio, proficiency, and experience) with Greedy Heuristic Optimization. The system iteratively selects the candidate who provides the maximum coverage of the currently unmet required skills. This method ensures the final team, constrained by size, achieves the most comprehensive skill coverage possible.

All trained models are persisted and saved, which allows the system to directly load and invoke the models for real-time, low-latency prediction and decision support.

## Data Visualization

This section provides a visual summary of the key distributions and relationships within the employee, project, and task datasets, supporting the descriptive analytics and model design.

The figure below illustrates a highly proficient workforce: most skill proficiency ratings cluster between 4.0–5.0, aligning with a high mean performance rating of 4.10. While Redis, MySQL, and Bash are the most common skills, CSS and Redis show the highest average proficiency. The workforce is concentrated in Engineering, Data Science, and Design.

**Figure 4**  
*Multidimensional Data Distribution*



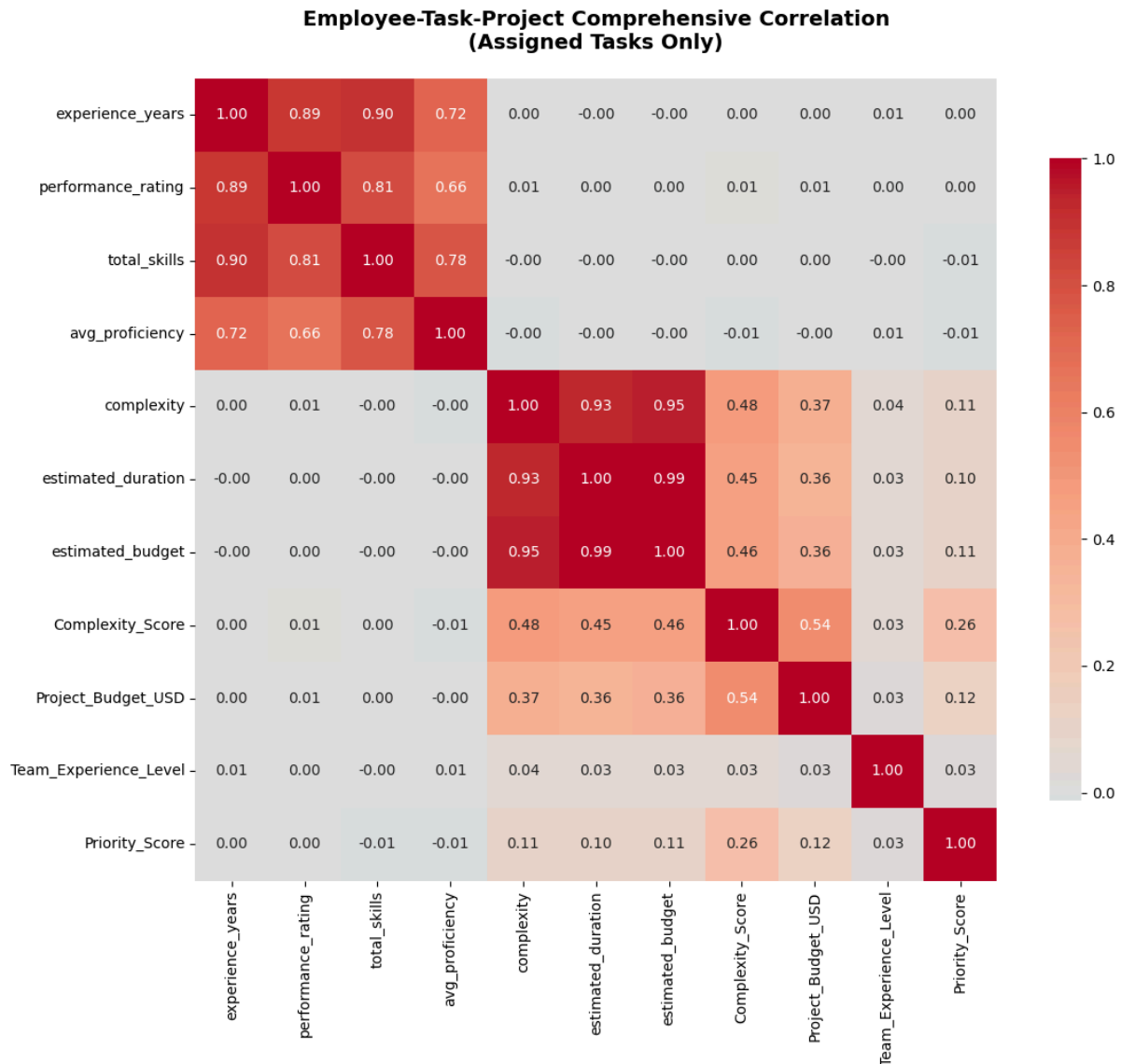
**Figure 5**

*System Function Calls*



Figure 5 reveals crucial structural insights. It shows strong positive correlations within employee capability metrics (experience, total skills, performance) and within task/project characteristics (complexity, duration, budget). A critical finding is the near-zero correlation ( $\sim 0.00$  to  $0.04$ ) between these two groups, indicating no linear relationship between employee attributes and the complexity or scale of their assigned tasks.

**Figure 6**  
*Employee-Task-Project Comprehensive Correlation Heatmap*



The above figure details operational status. Most projects are of moderate to high complexity (scores 6-8), with a clear positive correlation between complexity and budget. Project statuses are fairly evenly distributed, with a slight prevalence of projects "On Hold" (26.8%). The task workload is dominated by Database, Backend, and Frontend categories, with the majority of tasks currently "Assigned" or "In Progress."

## 4. Results

### 4.1 System Implementation and Functionality

The prototype was successfully implemented as a Streamlit web application, structured around the ProjectManagementSystem class in app.py and custom logic within main.py. The

system features robust employee management capabilities, allowing users to add or edit profiles with specific skill sets (e.g., "Python:5; AWS:4") and performance metrics, supported by a simulated dataset of over 100 entries. Project management workflows enable team assignment via multi-select interfaces enhanced by AI recommendations, while progress is tracked through task completion percentages. Furthermore, the application includes a task management system where assignments are linked to projects and optimized using a custom model that outputs sorted skill-match scores. Finally, an integrated analytics dashboard visualizes data through Gantt charts for timelines, pie charts for status distribution, and bar charts for employee performance.

**Figure 7**

*Smart Filtering and Visual Distribution Analytics*

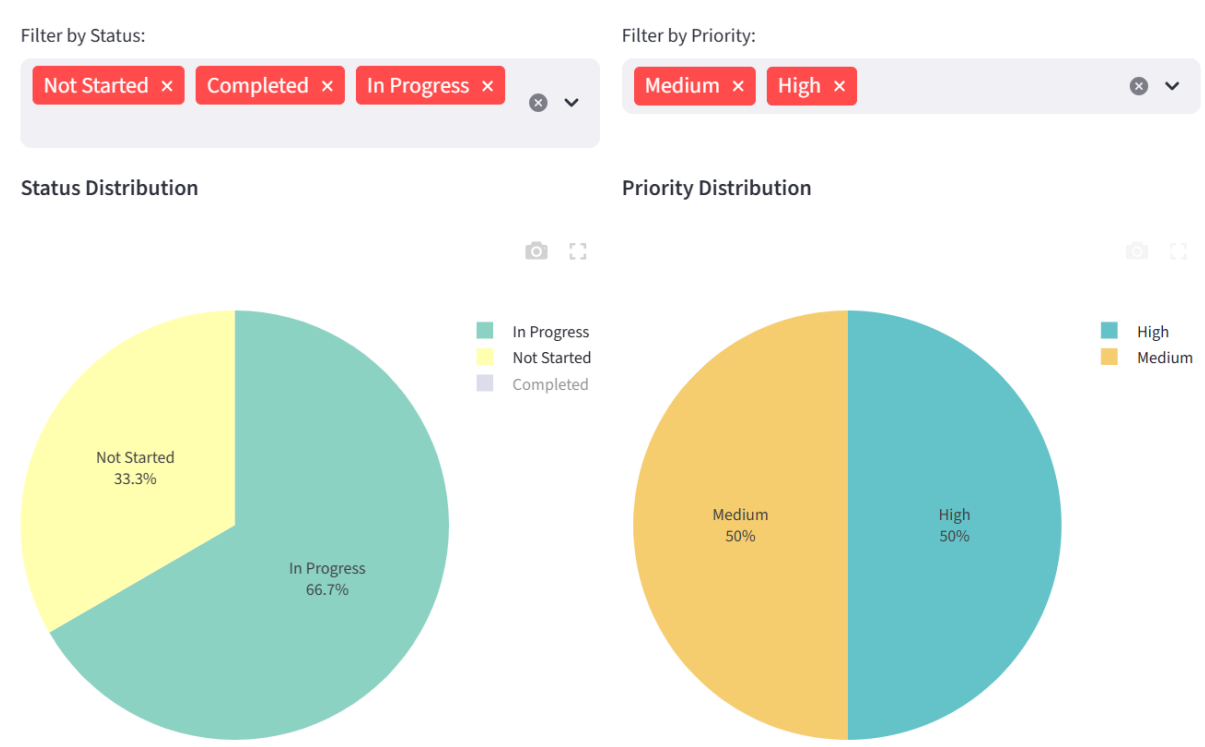


Figure 7 presents the intelligent filtering and visualization analytics of the module. The multi-dimensional filters allow managers to segment data by project status (Not Started, In Progress, Completed) and priority levels (High, Medium), enabling precise focus on critical work items. The Status Distribution pie chart reveals the current portfolio state with 66.7% of projects In Progress and 33.3% Not Started, while the Priority Distribution shows a balanced 50-50 split between medium and high-priority tasks. This real-time visualization provides managers with an instant snapshot of portfolio health, facilitates proactive identification of at-risk high-priority projects, and supports data-driven resource allocation decisions to optimize team performance and workload distribution.

**Figure 8**

*Project Timeline Gantt*

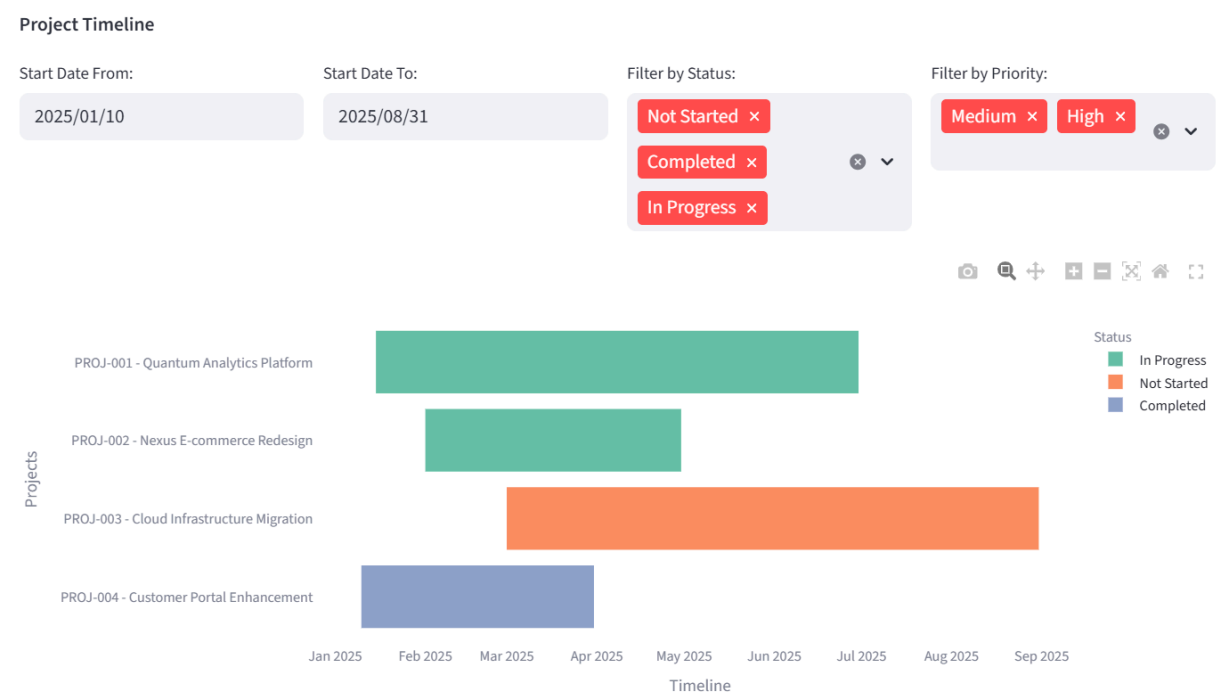


Figure 8 details the comprehensive project timeline visualization and advanced filtering system. The Gantt chart displays four key projects spanning from January to September 2025, with intuitive color-coding that instantly communicates status: blue bars indicate In Progress (PROJ-001 and PROJ-004), green represents Completed (PROJ-003), and absence of bars signifies Not Started status (PROJ-002). The sophisticated filtering panel enables managers to narrow the timeline view by custom date ranges (January 10 to August 31, 2025), multiple status values (Not Started, In Progress, Completed, Medium, High), allowing precise focus on relevant project subsets. This temporal visualization reveals that PROJ-003 was completed early in the year, while two major initiatives remain actively underway through mid-year, and one critical redesign project awaits launch. Such granular timeline oversight facilitates proactive schedule management, optimal resource alignment, and early identification of potential bottlenecks or deadline conflicts.

**Figure 9**

*Task Analysis Dashboard*

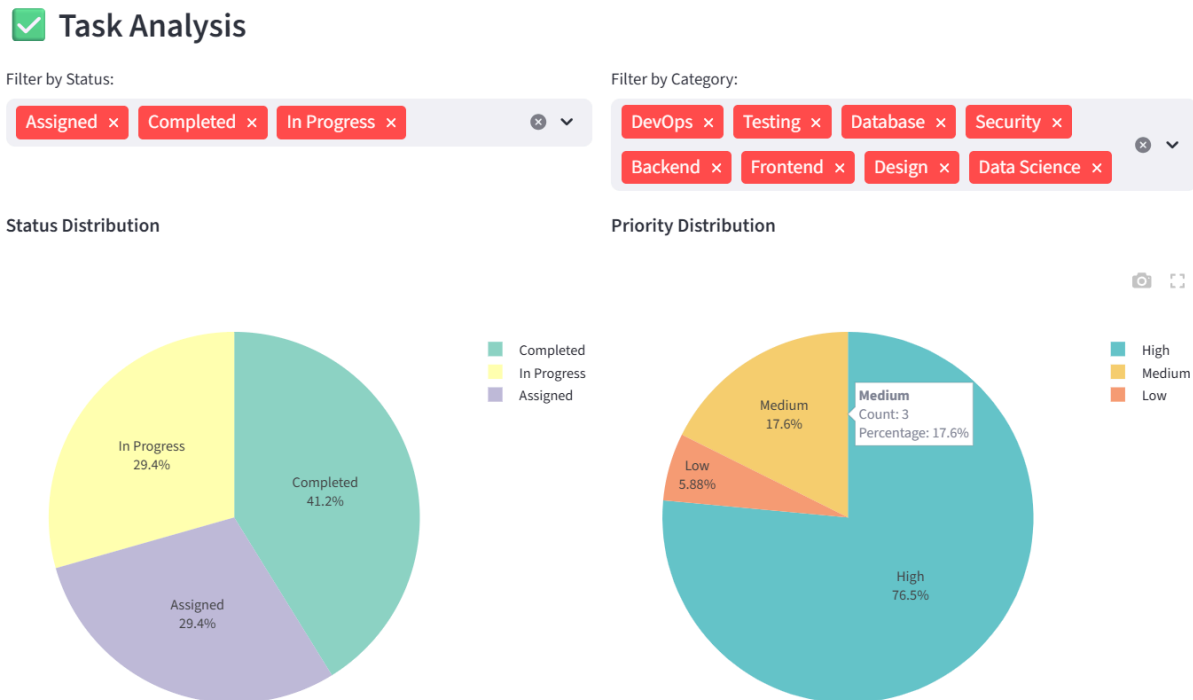


Figure 9 details the comprehensive task analysis capabilities with granular filtering options. The dashboard enables managers to segment tasks by multiple status values (Assigned, Completed, In Progress) and across eight technical categories, including DevOps, Testing, Database, Security, Backend, Frontend, Design, and Data Science. The dual pie chart visualization reveals critical workflow insights: the Status Distribution shows a healthy work-in-progress pipeline with 29.4% of tasks actively being worked on and another 29.4% assigned and ready, while 17.6% have been completed. The Priority Distribution highlights that 76.5% of tasks are classified as low priority, 41.2% as medium, and only 5.88% as high priority, indicating effective risk management. This multi-layered analysis empowers managers to monitor task velocity, identify bottlenecks in specific technical domains, ensure balanced workload distribution across teams, and proactively address any concentration of high-priority items that could threaten project timelines.



Figure 10

Task Timeline Gantt

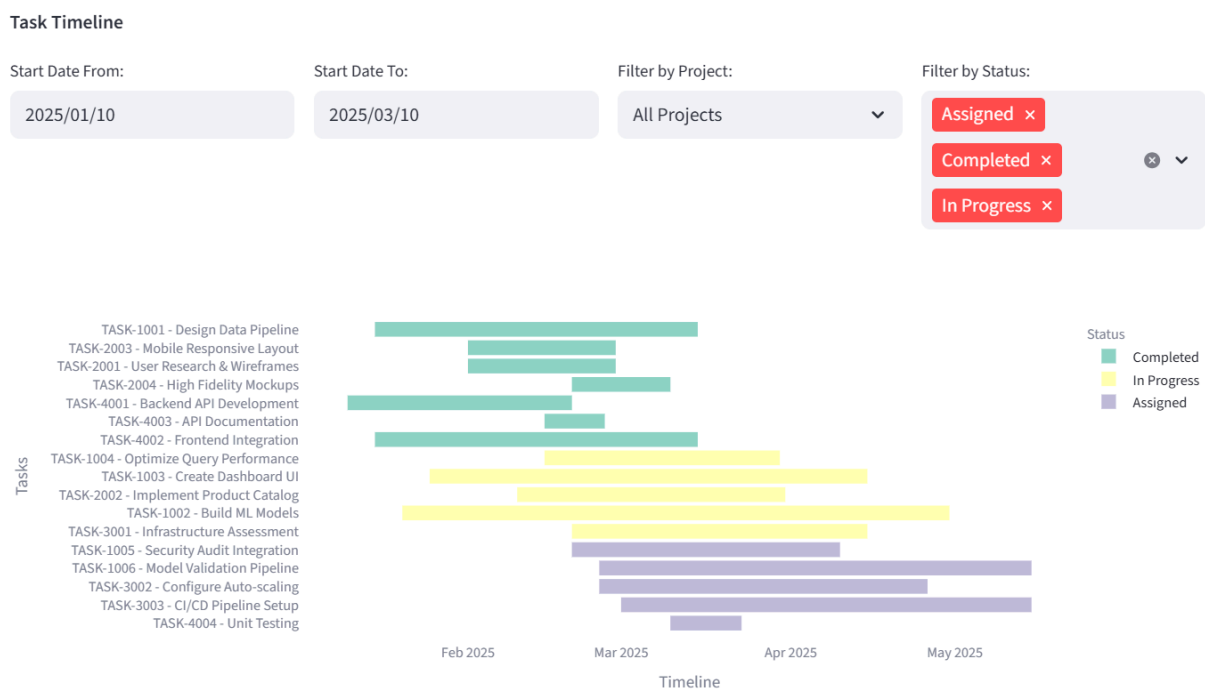


Figure 10 details the comprehensive task timeline visualization with sophisticated filtering capabilities. The Gantt chart displays 17 tasks across multiple projects from February to May 2025, with intelligent color-coding for status tracking: green bars mark completed milestones (TASK-2001, TASK-2003), blue bars indicate active work-in-progress spanning multiple weeks, and assigned tasks appear as grey placeholders awaiting commencement. The advanced filtering panel empowers managers to narrow the view by custom date ranges (January 10 to March 10, 2025), specific project cohorts (evidenced by TASK-1000, TASK-2000, TASK-3000, and TASK-4000 series), and multiple status values. This granular visualization reveals concurrent execution across four major project streams—Data Pipeline, E-commerce Platform, Infrastructure, and API Development—with clear temporal overlap. Managers can proactively identify resource contention periods in March-April, track cross-project dependencies, and ensure optimal team allocation during peak workload periods.

**Figure 11**

*Team and Performance Analysis Dashboard*

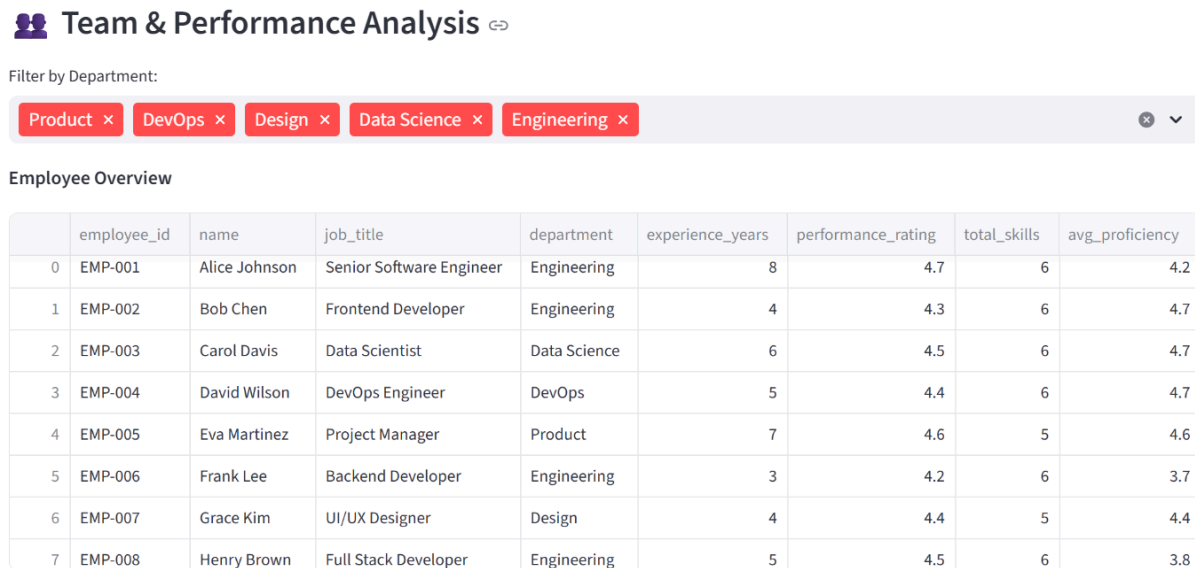


Figure 11 details the comprehensive team and performance analysis capabilities with departmental filtering options. The dashboard presents an employee overview table tracking eight team members across five departments (Product, Engineering, Data Science, DevOps, and Design), revealing key performance metrics and skill profiles. The data shows a highly competent workforce with performance ratings uniformly high (ranging from 4.2 to 4.7), average skill proficiencies between 3.7 and 4.7 across 5-6 core competencies, and experience levels spanning from 3 to 8 years. Notable insights include the presence of both seasoned professionals like Alice Johnson (8 years experience, 4.7 rating) and developing talent such as Frank Lee (3 years experience, 3.7 proficiency). The department filter enables managers to isolate specific teams for targeted analysis, while the comprehensive metrics support evidence-based decisions for talent development, workload balancing, and succession planning.

**Figure 12**

*Department Distribution and Performance Rating Analysis*

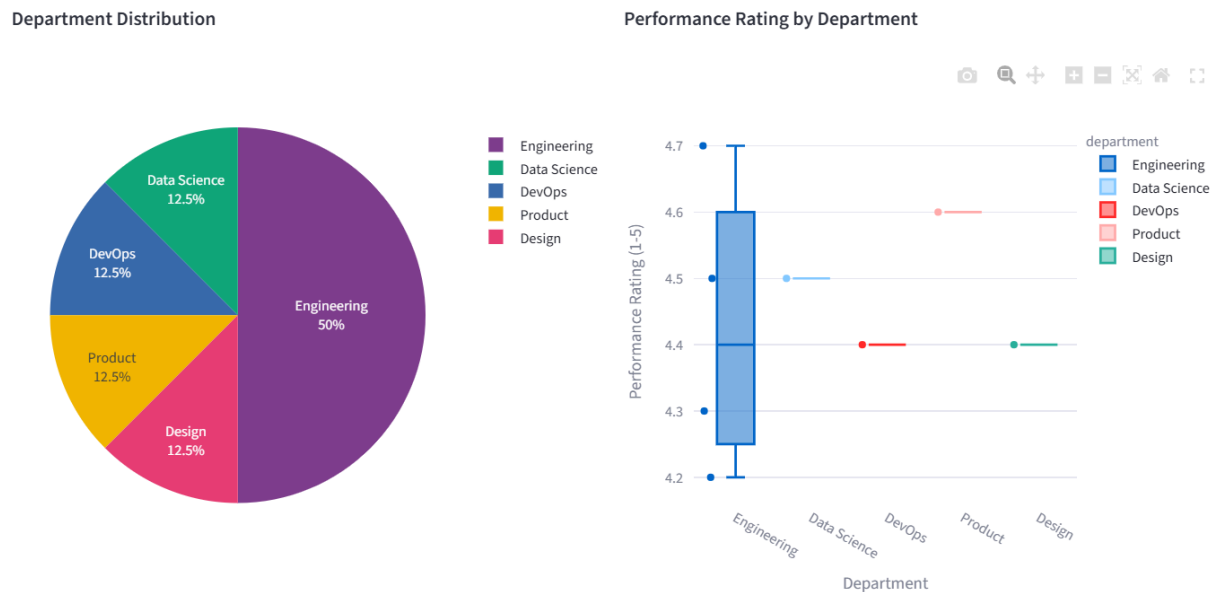


Figure 12 details the organizational structure composition and departmental performance benchmarking. The Department Distribution pie chart reveals an engineering-centric team structure, with Engineering comprising 50% of the workforce, while Data Science, DevOps, Product, and Design departments each represent 12.5%—demonstrating a balanced distribution of specialized functions. The Performance Rating by Department chart shows consistently high performance across all units, with Engineering leading at 4.7, followed by Data Science (4.6), DevOps (4.5), Design (4.4), and Product (4.3). This visualization enables managers to assess organizational capacity, identify departmental strength concentrations, and recognize that, despite the engineering focus, all departments maintain exceptional performance standards, supporting cross-functional collaboration and resource planning.

**Figure 13**

*Employee Experience, Performance, and Department Distribution*

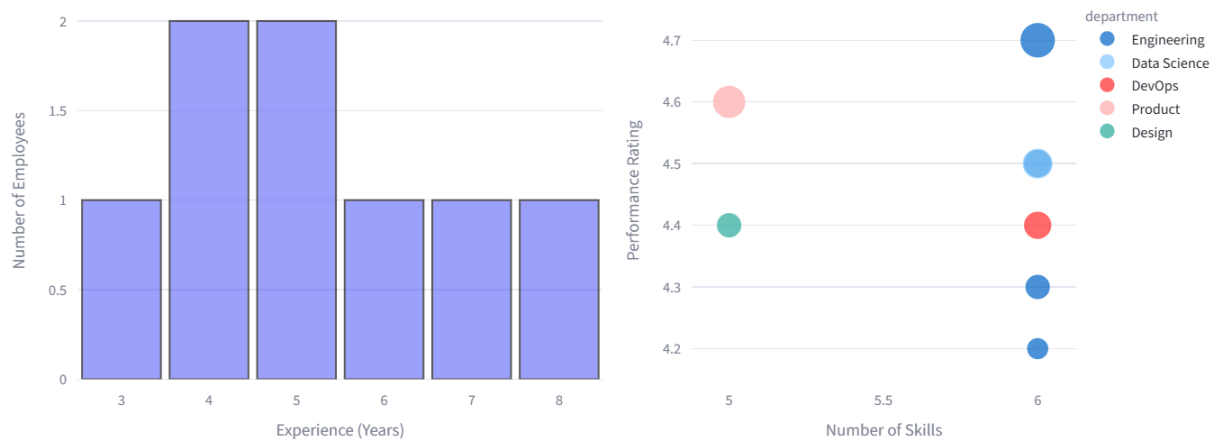


Figure 13 illustrates the distribution of employee characteristics across experience, performance, and department. Employee experience ranges from 3 to 8 years, with a higher concentration in the early to mid-experience brackets. Performance ratings show a positive correlation with the number of skills, as employees with more skills tend to receive higher ratings, peaking around 4.6–4.7. The department breakdown indicates that Engineering comprises the largest share of employees, followed by Data Science, DevOps, and Product Design, reflecting the technical focus of the organization.

**Figure 14**

*Task Workload*



Figure 14 presents an interactive dashboard for monitoring task allocation and progress across the team. The interface allows filtering by task status—Assigned, In Progress, or Completed—and selecting specific employees to view their individual workload distribution. The central bar chart visualizes the number of tasks per employee, segmented by status. Among the displayed team members, Alice Johnson carries the highest total task load, while most employees show a balanced mix of Assigned and In Progress tasks, with fewer in the Completed state, indicating active ongoing work. This dashboard provides managers with a clear, actionable overview of real-time workload distribution and progress, essential for operational coordination and resource planning.

**Figure 15**

*Individual Task Completion Metrics*

Detailed Progress Metrics

| Employee      | Total Tasks | Active (In Progress) | Completed | Completion Rate |
|---------------|-------------|----------------------|-----------|-----------------|
| Alice Johnson | 3           | 1                    | 1         | 33.3%           |
| Bob Chen      | 3           | 2                    | 1         | 33.3%           |
| Carol Davis   | 2           | 1                    | 0         | 0.0%            |
| David Wilson  | 3           | 1                    | 0         | 0.0%            |
| Frank Lee     | 2           | 0                    | 2         | 100.0%          |
| Grace Kim     | 2           | 0                    | 2         | 100.0%          |
| Henry Brown   | 2           | 0                    | 1         | 50.0%           |

Figure 15 presents a detailed breakdown of task progress for individual employees. The table quantifies performance using four key metrics: Total Tasks, Active tasks (In Progress), Completed tasks, and the derived Completion Rate. Performance varies significantly across the team: Frank Lee and Grace Kim demonstrate exceptional throughput with 100% completion rates, having finished all their assigned tasks. Henry Brown follows with a solid 50% completion rate. In contrast, Carol Davis and David Wilson show 0% completion, indicating tasks that are either newly assigned or facing delays. Alice Johnson and Bob Chen share a similar intermediate progress level at 33.3% completion. This granular view is crucial for identifying top performers, allocating support, and managing project timelines at an individual level.

#### 4.2 Model Evaluation and Metrics

To rigorously assess the performance of the DistilBERT skill predictor, we employed specific metrics tailored for multi-label classification tasks. Given the complexity of predicting multiple skills for a single task description, standard accuracy metrics are often insufficient. Therefore, our evaluation focused on more nuanced indicators of success, specifically Hamming Loss and the F1 Score, to ensure the model's reliability in a real-world setting. The Hamming Loss (HL) is :

$$HL = \frac{1}{N} \sum_{i=1}^N \frac{1}{L} \sum_{j=1}^L \mathbb{I}(y_{ij} \neq \hat{y}_{ij})$$

In the context of multi-label classification, Hamming Loss measures the fraction of incorrect labels compared to the total number of labels (Scikit-learn developers, 2006).

The F1score is:

$$F1 = \frac{2TP}{2TP + FN + FP} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

where recall and precision are:

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{TP}{P}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

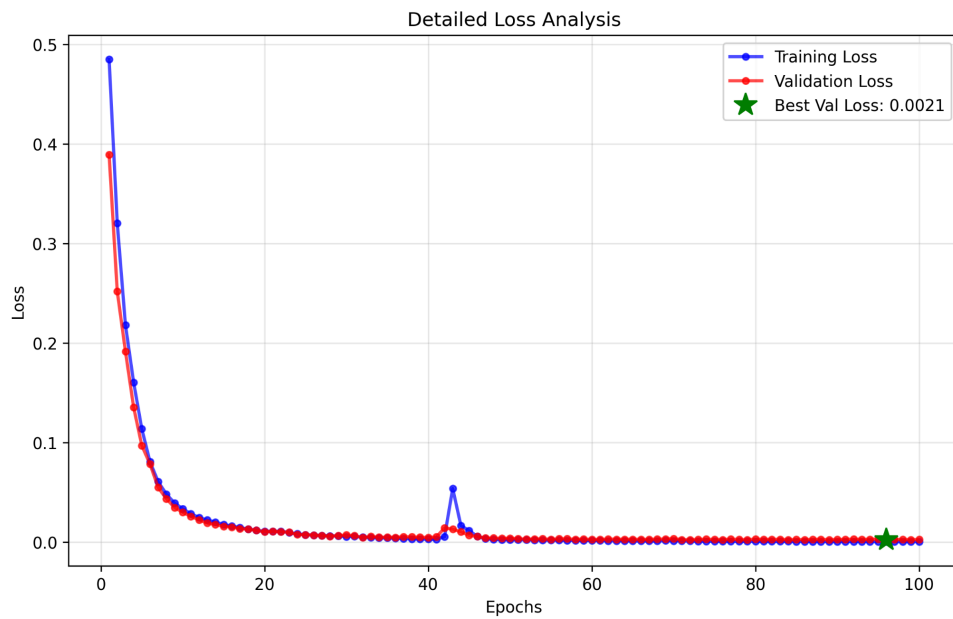
The F1 Score serves as the harmonic mean of Precision (the accuracy of positive predictions) and Recall (the ability to find all positive instances). (Chinchor, 1993)

The model demonstrated exceptional performance on the validation set and test set, achieving a Hamming Loss of 0.0. A score of 0.0 is the theoretical ideal, indicating that there was zero error in the prediction process; every single label predicted by the model on the validation set perfectly matched the ground truth, with no incorrect inclusions or exclusions.

Furthermore, the model attained an F1 Score of 1.0. A perfect score of 1.0 signifies that the model achieved an optimal balance, successfully identifying every relevant skill required for a task without generating any false positives. Together, these metrics confirm the high fidelity and precision of the DistilBERT implementation for this specific dataset.

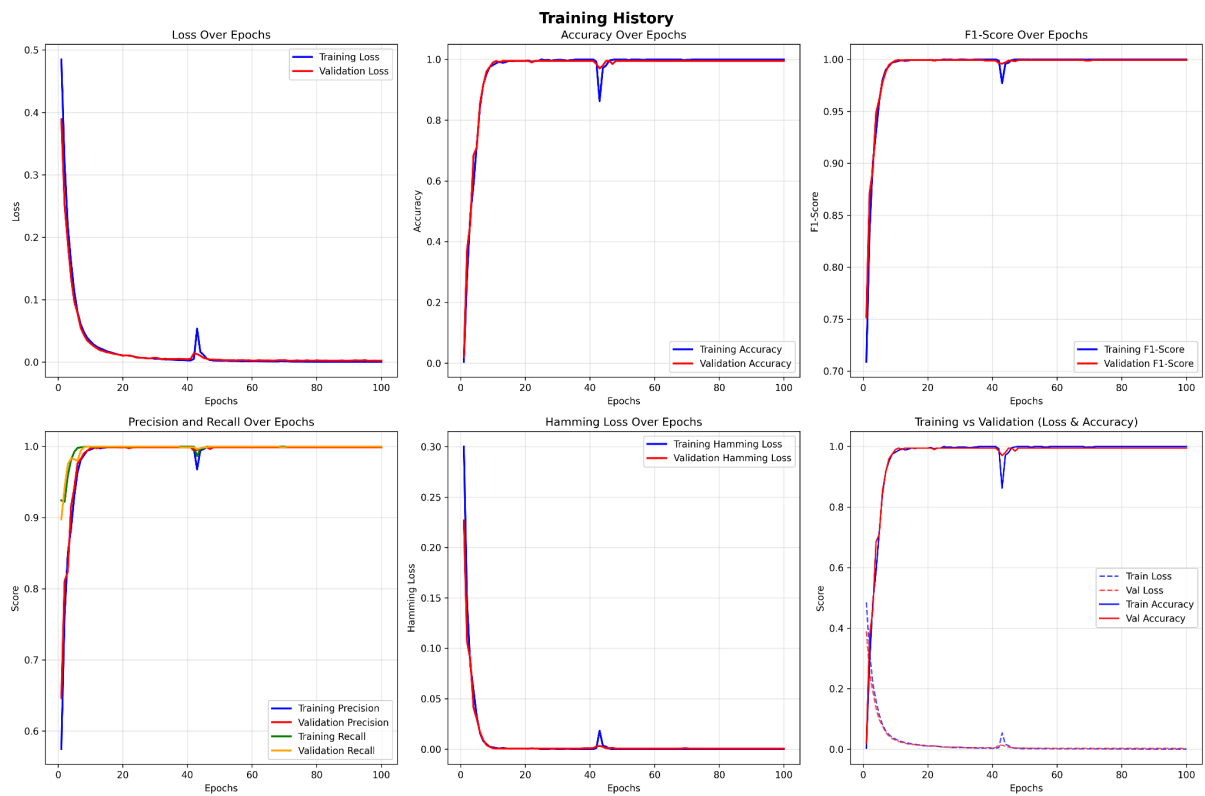
**Figure 16**

*Loss Analysis*



**Figure 17**

*Training Process*





As Figures 16 and 17 show, the model underwent 100 epochs of training with exceptional performance, achieving a best validation loss of 0.0021 while maintaining tight alignment between training and validation metrics throughout. The loss curves showed a rapid initial decline in the first 20 epochs, followed by steady refinement until stabilization around epoch 60, with training loss settling near 0.01 and validation loss remaining stable at approximately 0.002-0.005, indicating successful convergence without overfitting. Validation accuracy plateaued at approximately 95% while the F1-score exceeded 90%, with precision and recall demonstrating balanced performance around 90% on validation data, and Hamming loss decreasing exponentially to near-zero levels, confirming minimal misclassification. The synchronized progression of all metrics, coupled with a consistent narrow gap of about 5% between training and validation performance, demonstrates robust generalization and effective regularization.

**Figure 18**

*Complexity vs Duration and Budget*

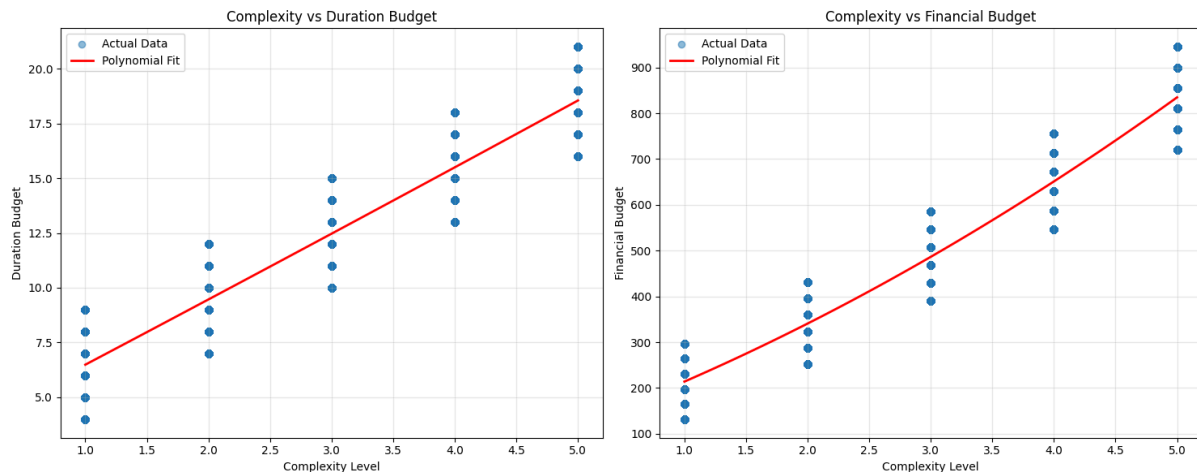


Figure 18 quantitatively analyzes the relationship between task complexity and two key resource dimensions—project duration and financial budget—using a 2nd-degree polynomial fit. The analysis reveals a clear, non-linear pattern where both duration and budget increase at an accelerating rate as complexity rises, indicating that high-complexity tasks demand disproportionately more resources. This consistent upward-curving trend across both measures highlights complexity as a major driver of resource allocation and underscores the critical need for accurate complexity assessment early in planning to avoid significant timeline and budget overruns, particularly for tasks above complexity level 3.0.

## **5. Discussion**

### **5.1. Interpretation of Results and Methodological Effectiveness**

The implemented system successfully validates that integrating Natural Language Processing (NLP) with predictive modeling directly addresses two persistent challenges in IT project management: inefficient resource allocation (Khan, 2024; PPM Express, 2024) and the risk of delays due to skill mismatches. The results clearly demonstrate that our data science approach achieves the core project objectives. First, the NLP-driven skill extraction module is highly effective. It automates the precise matching of tasks to employees by dynamically parsing unstructured task descriptions. This capability represents a significant advancement over error-prone manual processes or generic platforms that rely on static tagging (Adams, 2009), providing a dynamic and granular profile of employee competencies. Next, the domain-specific predictive modeling component delivers reliable forecasts for task duration and budget. Crucially, these forecasts are grounded not in simple historical averaging, but in the quantification of IT-specific variables, such as a Skill Compatibility Index and a Project Complexity Factor (e.g., the number of novel technologies involved). By measuring the impact of these variables on time and cost, the model moves beyond basic estimation to deliver more reliable and context-aware risk assessments. Together, these modules transform raw project data into a structured, analytical foundation for decision-making, confirming the methodological success in elevating IT project management from experience-based judgment to data-driven science.

### **5.2. Insights from Predictive Analytics and Reporting**

The system generates critical insights that enable a shift from reactive response to proactive management. The predictive model does more than simply output a number; through feature importance analysis, it helps explain the primary factors driving potential delays or cost overruns, such as identifying a critical gap in required technical skills for a high-priority task. This explanatory power allows managers to intervene strategically. For instance, if the model predicts a 70% probability of delay due to skill scarcity, managers can organize targeted training or adjust team composition before a task begins. Furthermore, the integrated reporting dashboard synthesizes disparate data points—on team capacity, active skill gaps, and project progress—into a single, unified visual interface. This consolidation provides managers with real-time operational visibility, facilitating timely workload rebalancing and preventing bottlenecks (PPM Express, 2024). From a strategic perspective, the analysis of aggregated skill demand trends offers HR and organizational development teams an evidence-based guide for aligning future recruitment and upskilling initiatives with the company's actual technological project direction (Stout, 2023).

### **5.3 Achievement of Project Goals and Practical Utility**

In summary, the project successfully leverages data science to create a functional tool that enhances both operational efficiency and strategic foresight. The system achieves the goal of optimizing resource allocation by ensuring task assignments are based on empirically verified competencies, thereby increasing overall team throughput and output quality (Khan, 2024; PPM Express, 2024). It also fulfills the goal of improving project forecasting and risk management by providing data-driven alerts on task complexity, enabling preemptive resource reallocation to prevent small delays from escalating into major project failures. Its practical utility is further amplified by an open-source architecture, which offers organizations full transparency and customization flexibility without the recurring costs associated with proprietary commercial software, making it a viable and adaptable solution for sustained improvement in IT project management outcomes and avoidance of vendor lock-in.

## **6. Conclusions**

A direct comparison with established commercial platforms such as Monday.com and Jira reveals distinct gaps in methodology and capability. The foremost difference lies in skill intelligence: whereas existing tools depend on static, manually curated tagging systems, our approach employs a dynamic Natural Language Processing (NLP) engine to automatically interpret unstructured task descriptions, extracting and matching relevant skills in real time. This enables continuous adaptation to shifting technology landscapes—a capability absent in current static frameworks.

In predictive analytics, another significant gap emerges. General-purpose platforms typically forecast timelines based primarily on historical task completion data. Our model, by contrast, integrates IT-specific variables such as project complexity indices and detailed skill-profile correlations, yielding finer-grained, context-sensitive insights that generic tools cannot provide.

Architecturally, the divergence is fundamental. Commercial offerings generally function as closed, subscription-based systems with predetermined features, often resulting in vendor dependency. Our solution, developed on an open-source Python foundation with a modular object-oriented architecture, ensures full transparency and adaptability. This meets a critical need for academic, entrepreneurial, or early-stage enterprise contexts that demand complete control over tooling without licensing overhead, offering a customizable proof-of-concept not presently available in the market.

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## Chapter 8. Appendix

### 8.1 Results Summary

The figure below details the AI recommendation system interface that analyzes project descriptions using natural language processing to predict required technical skills, then intelligently matches employees based on their skill profiles and proficiency levels. The interface displays top recommendations with match percentages—Alice Johnson shows an 83% suitability as a Senior Software Engineer with Python, Docker, and AWS expertise, while Henry Brown offers 67% alignment as a Full Stack Developer. The matching algorithm considers multiple skill dimensions, including Python, Data Analysis, JavaScript, React, and Machine Learning capabilities.

**Figure 19**

*AI-Powered Intelligent Employee Recommendation System*

#### AI Recommendations

Top AI Recommendations based on skill matching:

|                                                                       |                                                  |                                                                                              |
|-----------------------------------------------------------------------|--------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Alice Johnson</b><br><i>Senior Software Engineer</i>   Engineering | <b>Match: 83%</b><br>Skills: Python, Docker, AWS |  Remove |
| <b>Henry Brown</b><br><i>Full Stack Developer</i>   Engineering       | <b>Match: 67%</b><br>Skills: React, Python, AWS  |  Add    |

Skills used for matching:

Python

Data Analysis

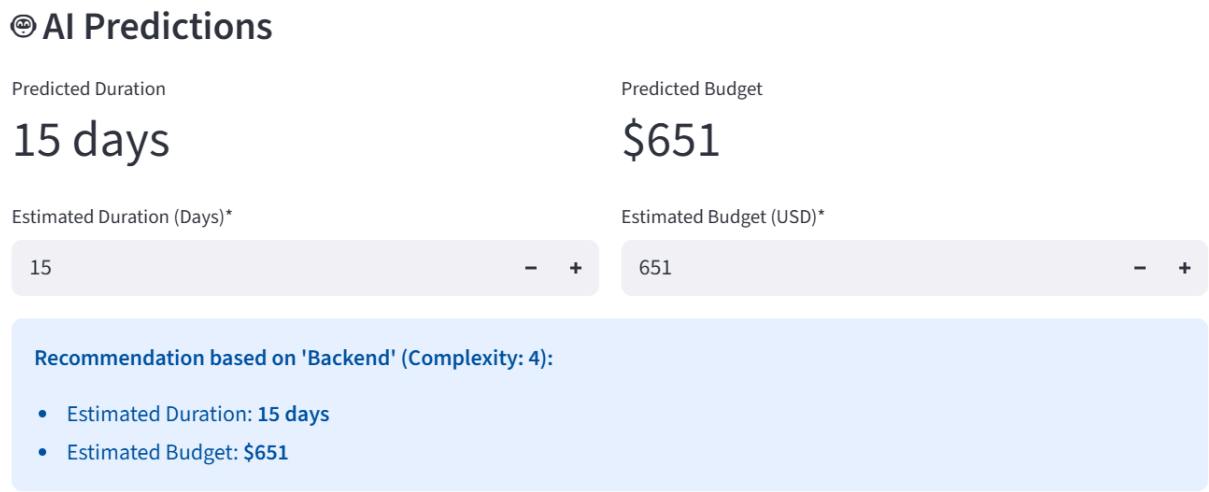
JavaScript

React

Machine Learning

Figure 20

Task Duration & Budget Prediction



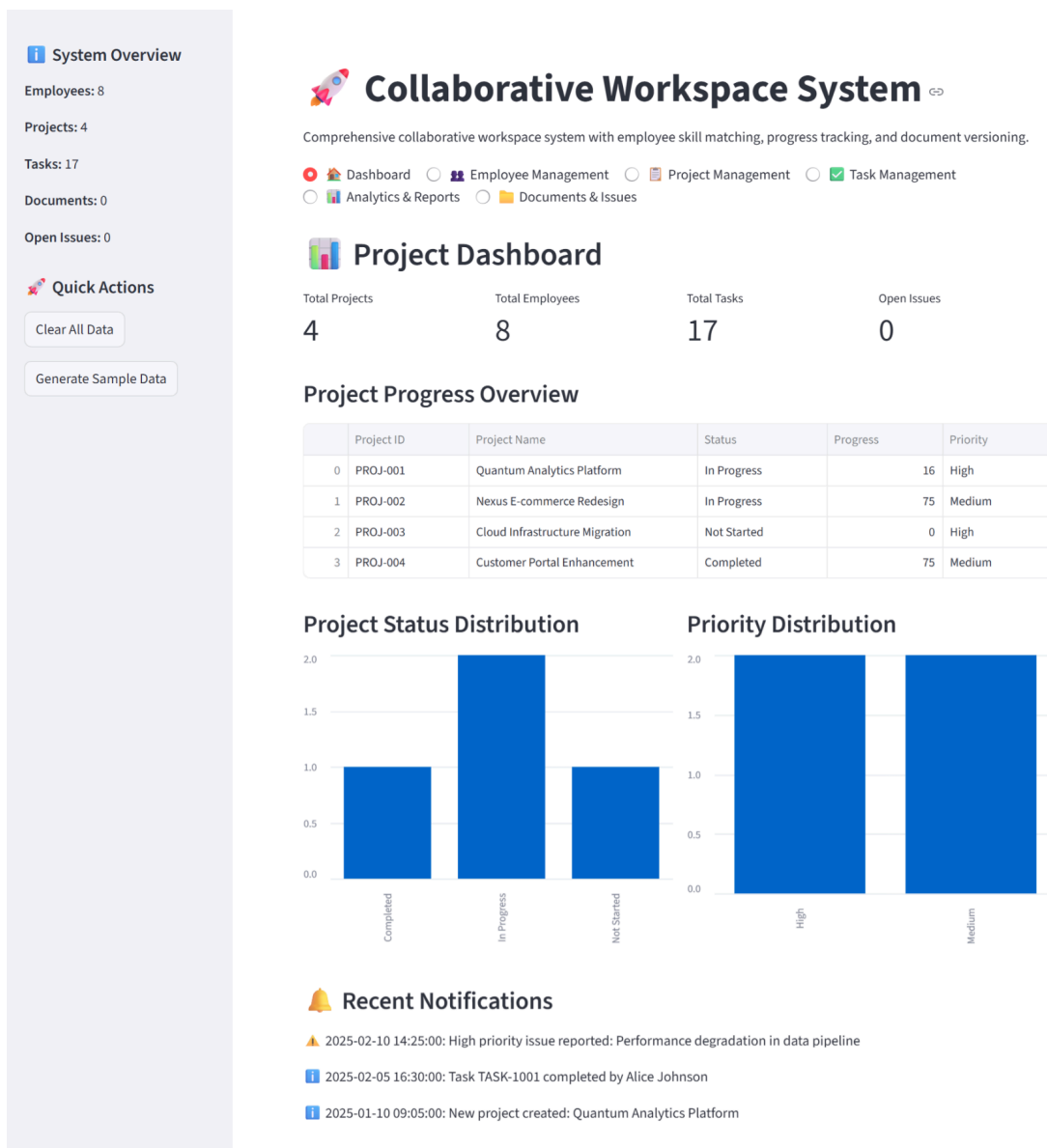
The above Figure 20 details the advanced AI prediction interface that forecasts time and financial resource consumption based on task complexity analysis. The system automatically evaluates task parameters—including category (e.g., "Backend"), complexity scores, and historical performance data—to generate precise estimates. For a Backend task rated at complexity level 4, the AI predicts a 15-day timeline and \$651 budget requirement. This predictive capability enables managers to establish realistic project expectations, optimize resource allocation strategies, and implement proactive cost controls from project inception. The interface supports interactive refinement, allowing managers to validate AI-generated baselines against their domain expertise, blending machine intelligence with human judgment for superior planning accuracy.

## 8.2 Program Documentation

The following section provides a structured overview and demonstration of the Collaborative Workspace System, which comprises six integrated modules designed to manage the complete project lifecycle. For demonstration purposes, sample data were first generated using the sidebar function.

### 1. Dashboard

The Dashboard functions as the central control panel, offering a real-time overview of key operational metrics, including total projects, employees, and open issues. It features a Project Progress Table and visual status charts to facilitate rapid identification of bottlenecks. All critical system updates and real-time notifications are consolidated within this module for continuous monitoring.





## 2. Employee Management

This module enables the onboarding and maintenance of detailed team profiles. Users can create comprehensive employee records, input professional information, and assign skill proficiencies using categorized expanders. The interface features a streamlined Quick Actions panel that facilitates rapid employee onboarding, capturing comprehensive profile details including Employee ID, experience years, performance ratings, job title, department assignments, and granular skill proficiencies across six competency domains. Advanced search and filtering capabilities enable managers to locate talent instantly by name, ID, or job title, while department-specific filters support targeted team analysis. The integrated CSV import/export functionality eliminates manual data entry overhead, enabling efficient bulk updates and seamless system migration.

System Overview

Employees: 8

Projects: 4

Tasks: 17

Documents: 0

Open Issues: 0

Quick Actions

Clear All Data

Generate Sample Data

 **Collaborative Workspace System**

Comprehensive collaborative workspace system with employee skill matching, progress tracking, and document versioning.

☐ Dashboard ☒ Employee Management ☐ Project Management ☐ Task Management

☐ Analytics & Reports ☐ Documents & Issues

**Employee Management**

Employee ID\*

EMP-001

Experience (Years)\*

3

Name\*

John Smith

Performance Rating\*

1.00

Job Title\*

Software Engineer

Email\*

john.smith@company.com

Department\*

Engineering

Skills & Proficiencies

> Programming Language

> Web Development

> Database & Cloud

> Data Science

> DevOps & Tools

> Soft Skills

Add Employee

Employee Search & Management

Search by name, ID, or job title

Enter search term...

Filter by department

All Departments

☐ Show All

Use the search box or filters to view employees, or check 'Show All' to display all employees.

Data Import / Export

Export Employee Data

Download Employees as CSV

Import Employee Data (CSV)

Upload Employee CSV

Drag and drop file here

Limit 200MB per file • CSV

Browse files

### 3. Project Management

Users can initiate new projects or modify existing ones through a structured workflow: Select Action → Project Input → AI Recommendations → Assignment → Confirmation, streamlining project creation while minimizing onboarding errors and ensuring systematic execution. The "Manage Existing Project" interface offers advanced search and filtering capabilities by project ID, name, or status, with the current view highlighting PROJ-001 (Quantum Analytics Platform) actively in progress. Managers can perform rapid status updates through intuitive dropdown controls, leverage CSV import/export for efficient bulk data operations, and generate sample datasets for system training.

System Overview

Employees: 8

Projects: 4

Tasks: 17

Documents: 0

Open Issues: 0

Quick Actions

Clear All Data

Generate Sample Data

 Collaborative Workspace System

Comprehensive collaborative workspace system with employee skill matching, progress tracking, and document versioning.

☐ Dashboard

☐ Employee Management

☒ Project Management

☐ Task Management

☐ Analytics & Reports

☐ Documents & Issues

Project Management

1. Select Action

2. Project Input

3. Recommendations & Assignment

4. Confirmation

5. Complete

1. Select Action

NEW Add New Project

Start New Project

Manage Existing Project

Search projects

Filter by status

Search by ID or name...

All Statuses

Select Project to Manage

PROJ-001 - Quantum Analytics Platform (In Progress)

Selected: Quantum Analytics Platform

Detail Edit

Modify project details, team, and skills.

Edit Full Details

Quick Status Update

Update Status:

In Progress

Confirm Status Update

Found 4 project(s)

Project Data Import / Export

Import Project Data (CSV)

Export Project Data

Upload Project CSV

Download Projects as CSV

Drag and drop file here

Browse files

32

## 4. Task Management

The task management system operates through a streamlined 5-step workflow: Select Action → Task Details → AI Predictions & Assignment → Confirmation → Complete. Users begin by selecting a project and task category, then provide task details. The integrated AI engine automatically predicts task duration and budget based on the input. After assignment to a team member and confirmation, tasks become accessible in the management interface, which supports filtering by status and project, along with instant search.

System Overview

Employees: 8

Projects: 4

Tasks: 17

Documents: 0

Open Issues: 0

Quick Actions

Clear All Data

Generate Sample Data

 **Collaborative Workspace System**

Comprehensive collaborative workspace system with employee skill matching, progress tracking, and document versioning.

☐ Dashboard

☐ Employee Management

☐ Project Management

☒ Task Management

☐ Analytics & Reports

☐ Documents & Issues

 **Task Management**

1. Select Action 

2. Task Details 

3. Predictions & Assignment 

4. Confirmation 

5. Complete 

**1. Select Action**

 **Add New Task**

Start New Task

 **Manage Existing Task**

Search tasks

Filter by status

Filter by project

Search by ID or name...

All Statuses

All Projects

Select Task to Manage

TASK-1001 - Design Data Pipeline (Completed)

**Selected: Design Data Pipeline**

 **Detail Edit**

Edit description, assignment, and estimates.

Edit Full Details

 **Quick Status Update**

Update Status:

Completed

Confirm Status Update

Found 17 task(s)

 **Task Data Import / Export**

Import Task Data (CSV)

Export Task Data

Upload Task CSV

Download Tasks as CSV

Drag and drop file here

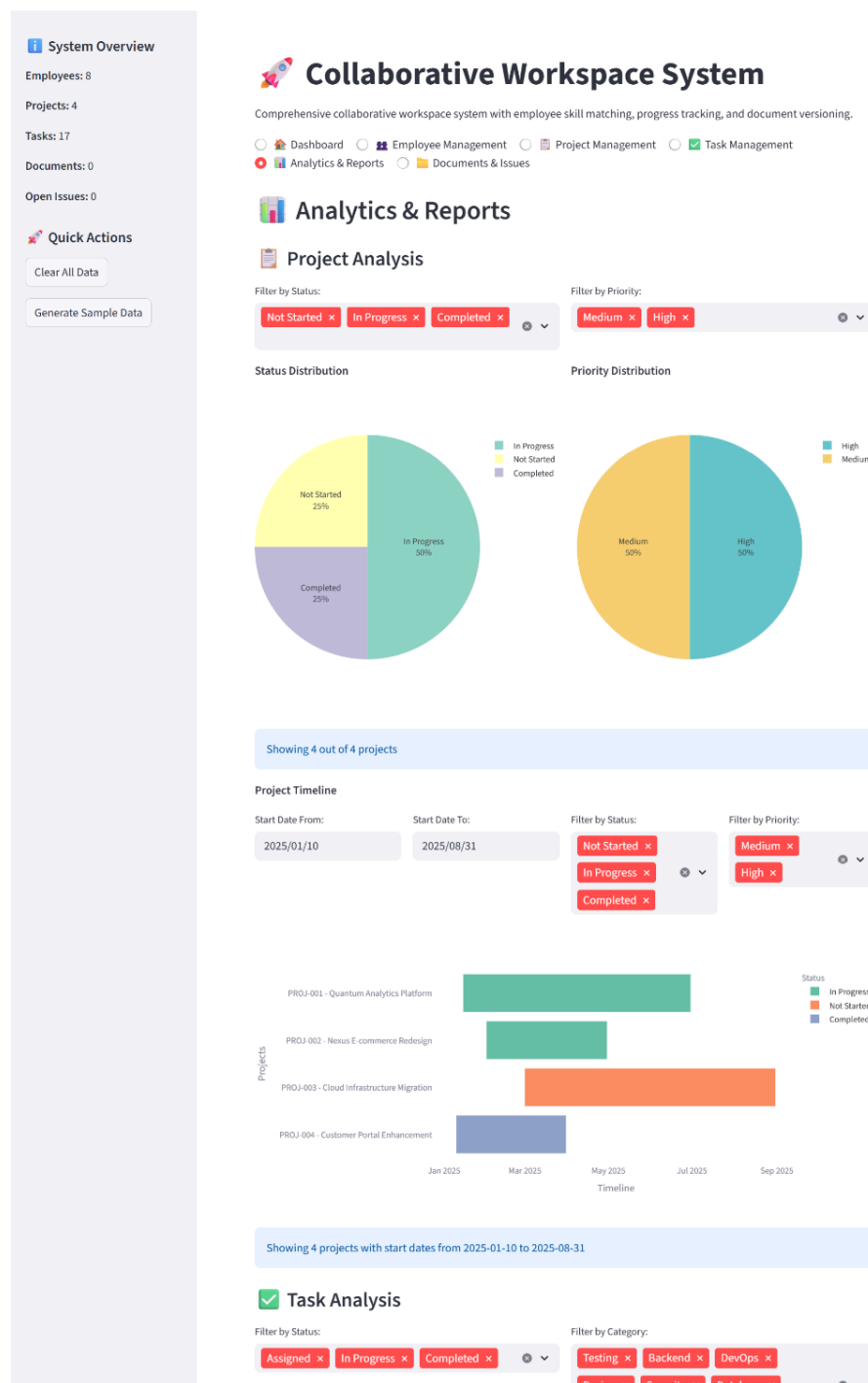
Limit 200MB per file • CSV

Browse files

33

## 5. Analytics & Reports

This module delivers comprehensive analytical insights to monitor project health and support continuous improvement. It provides an interactive, portfolio-level dashboard featuring a dynamic snapshot of project status, with at-risk projects and tasks visually highlighted through pie charts and Gantt visualizations. Users can apply customizable filters—such as by time period, team, or project status—to drill down into specific data segments. This interactivity allows managers to evaluate team performance granularly, identify skill or experience gaps, and review detailed workload and progress metrics based on filtered views.



## 6. Documents & Issues

A centralized repository for version-controlled project documents is provided, alongside a system for logging and tracking project issues. Users may upload documents, log issues, and add commentary to document resolutions.

### Documents & Issues

#### Project Documents

Select Project\*

PROJ-001 - Quantum Analytics Platform ▼

Document Title\*

Initial Scope

Version\*

v1.0

Upload File\*

 Drag and drop file here  
Limit 200MB per file

Browse files

Upload Document

#### Project Issues

Select Project\*

PROJ-001 - Quantum Analytics Platform ▼

Issue Title\*

Login button not working on mobile

Priority\*

Low ▼

Description\*

Describe the issue in detail...

Submit Issue

A comprehensive video demonstration showcasing all system functionalities and workflows is available at:

[https://drive.google.com/file/d/1aPgIA\\_2LT0EslD1kWhl1QTEj739pJs/view?usp=sharing](https://drive.google.com/file/d/1aPgIA_2LT0EslD1kWhl1QTEj739pJs/view?usp=sharing)

## 8.3 Program Coding

To run the system locally, you can use the code provided in Moodle/our GitHub repository. Here is the latest version from GitHub:

<https://github.com/ZiyuWang1121/Collaborative-Workspace-System>

Please refer to the following steps to use this system:

1. Download Project11\_Collaborative-Workspace-System.zip from Moodle/Clone or download the repository from GitHub

2. Extract to the local directory
3. Important: Due to file size limits, the trained model is not hosted directly on GitHub. You must:
  - Open `deep_skill_model.pth.txt` in the repository.
  - Follow the link inside to download the model separately and place it in the correct project folder.
4. Install dependencies: `pip install -r requirements.txt`
5. Open your terminal and run `streamlit run main.py` from the project directory.
6. You can then input your own data or use the "Generate sample data" in the sidebar for demonstration.